



Szilícium layouttól a kapcsolási rajzig

Beregnyei Balázs

bereg@impulzus.com

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Főmenü

- A 6502-es visszafejtése, motivációk, módszerek
- Tranzisztoros kapcsolások, a víz analógia
- A layoutfejtés grafikus szabályai

Reverse engineering motivációk

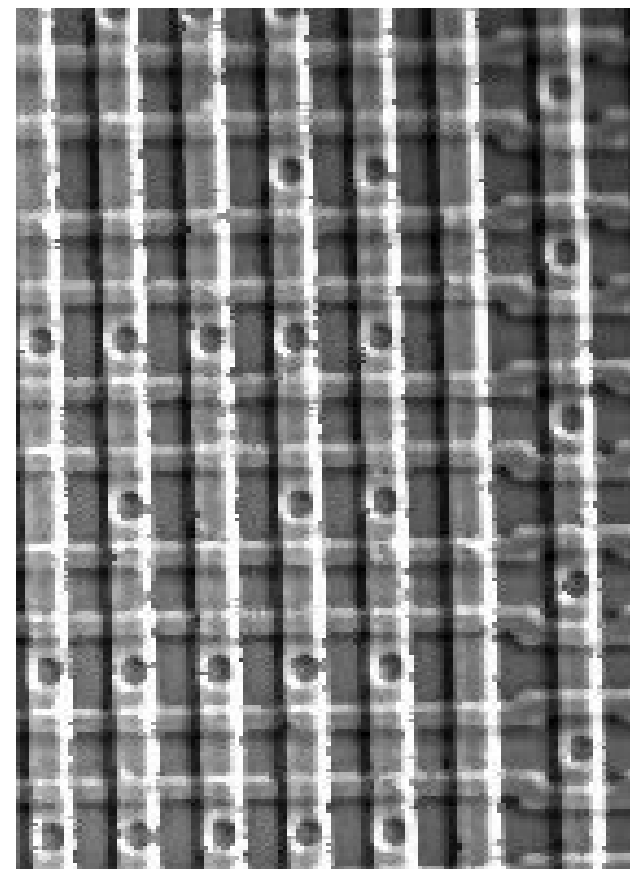
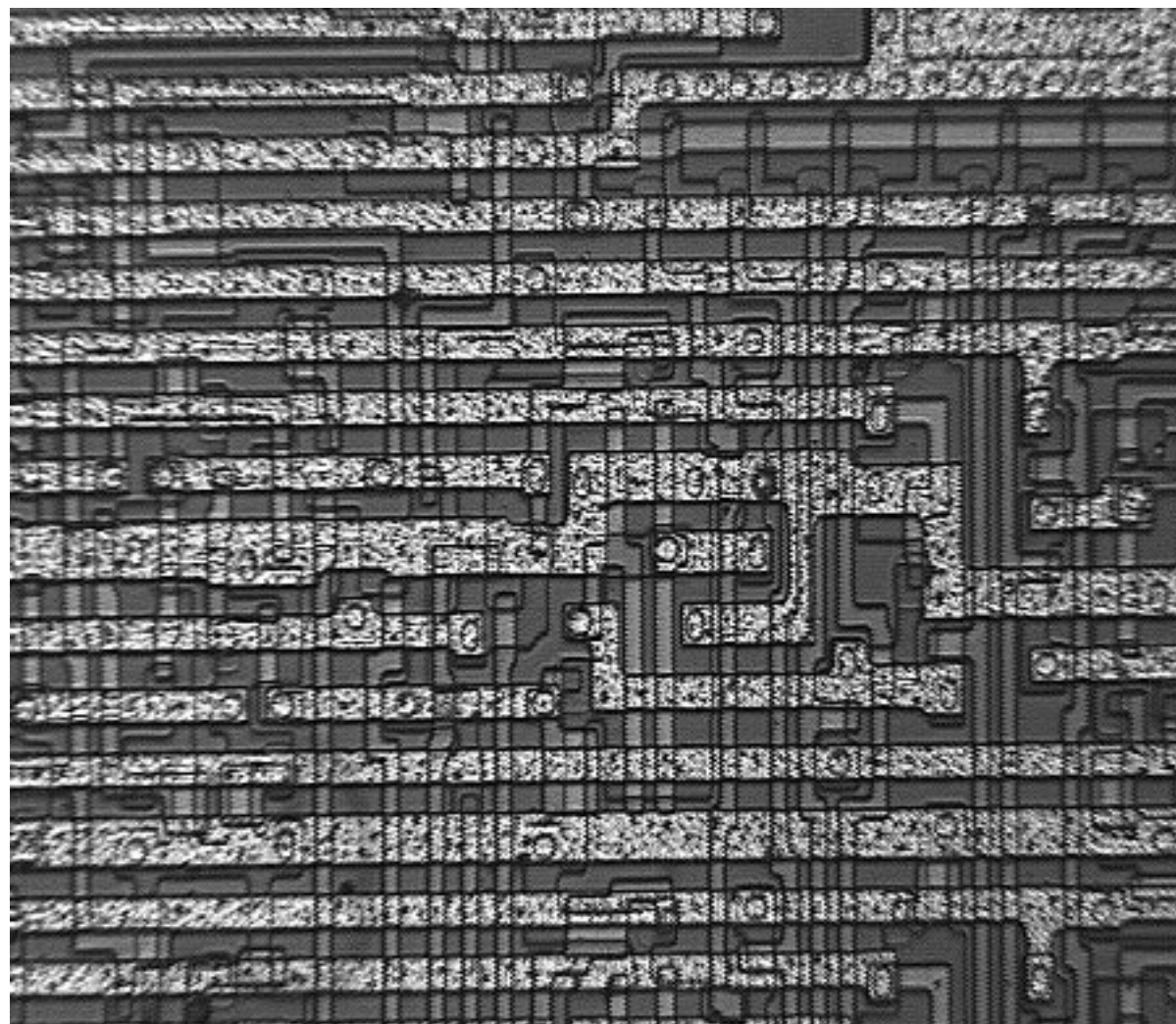
- Kódvédelem feltörése mikrokontrollereken
- Dokumentációs problémák megoldása
- Oktatási célok

Miért éppen a 6502?

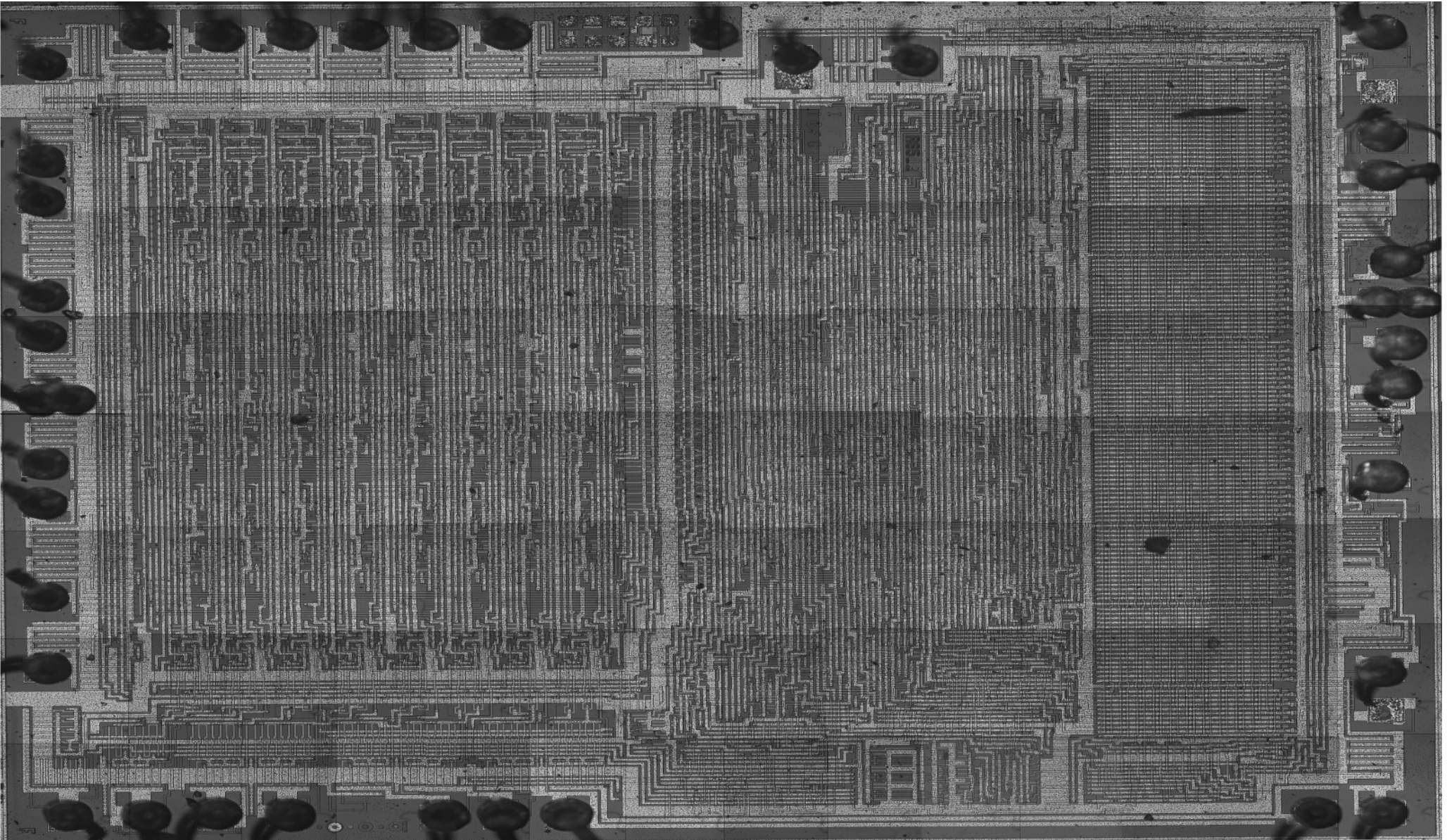
- 6 mikronos technológia, optikai mikroszkóppal könnyen fejthető
- Commodore korszak
- Nemdokumentált műveleti kódok: “tiltott kódok”



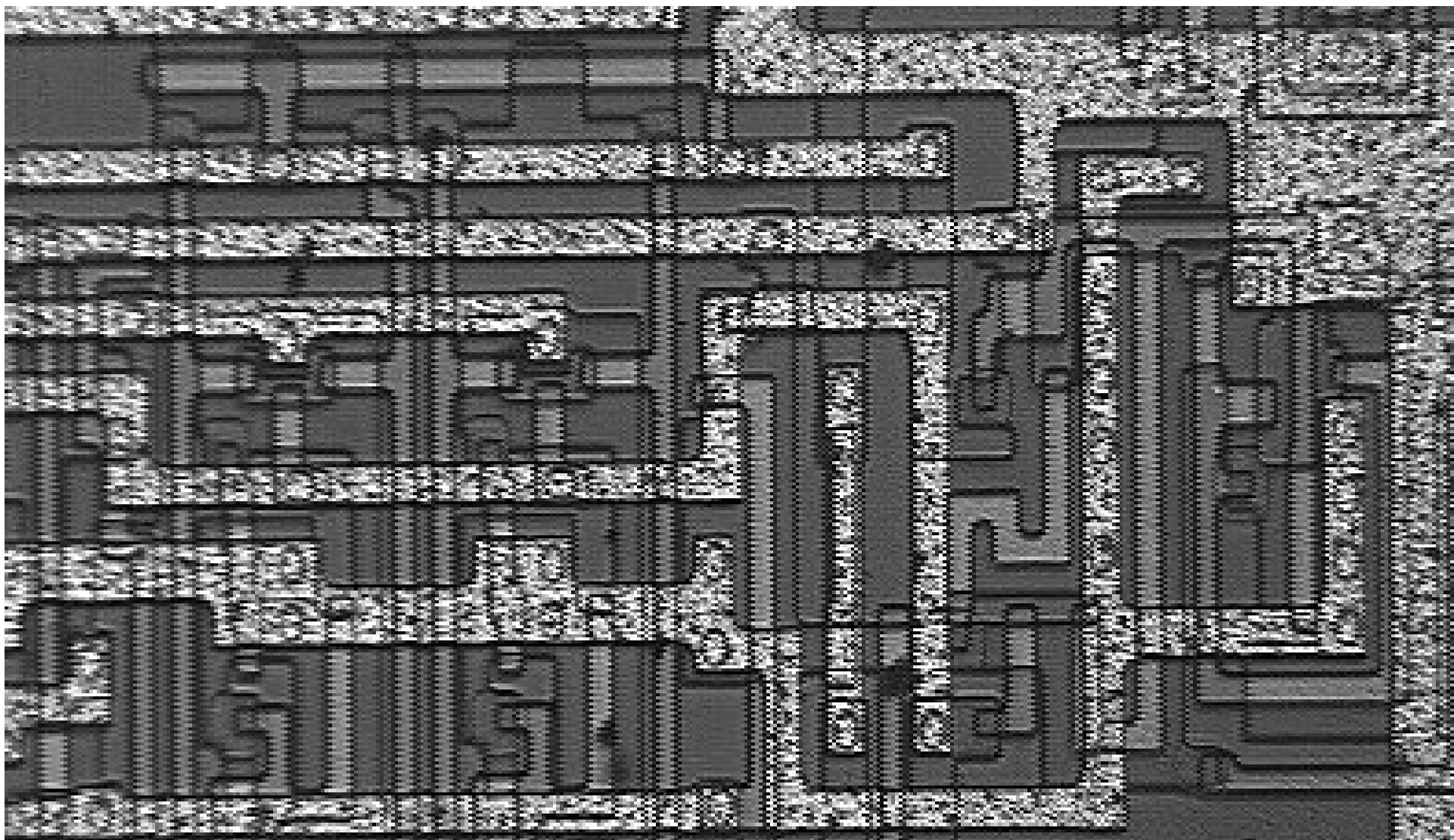
A fejtési munka kezdetei



Illesztés



Szabálykészlet?



- Csak az utaitásdekódoló ROM megfejtéséhez
használtam célszoftvert
- Kézi fejtés
- Kapcsolási rajz rendezése

The Visual 6502

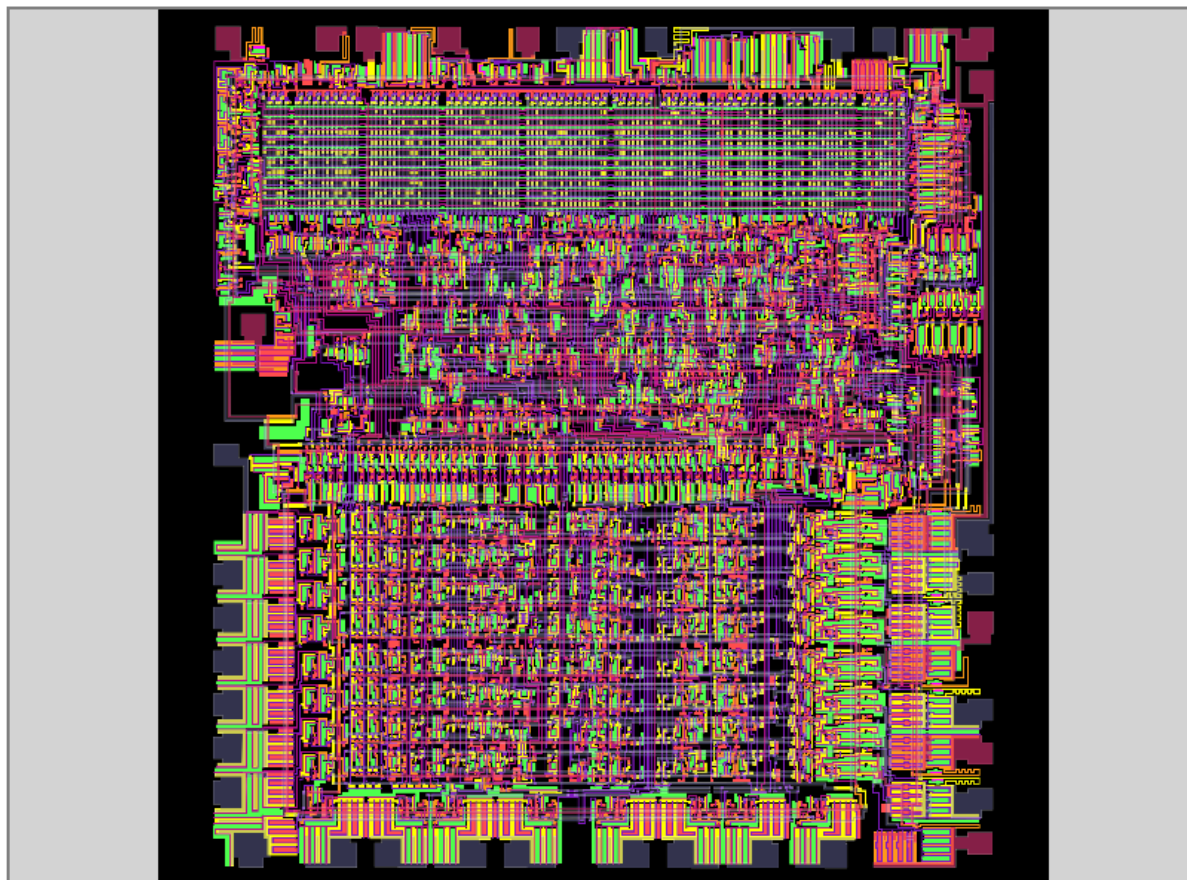
[FAQ](#) [Blog](#) [Links](#)

This simulator uses HTML5 features only found on the latest versions of browsers and needs lots of RAM. If you have trouble, please [check compatibility](#).

Keyboard controls: 'z' to zoom in, 'x' to zoom out, 'n' to step the simulation.

Mouse controls: Left-click and drag to scroll around (when you're zoomed in.)

More information in the [User Guide](#).



⏪ ⏴ ⏵ ⏩ ... or try [Advanced](#)

```
halfcyc:0 phi0:0 AB:0000 D:a9 Rnw:1
PC:0000 A:aa X:00 Y:00 SP:fd nv-BdIZc
Hz: 1 0
0000: a9 00 20 10 00 4c 02 00 00 00 00 00 00 00 00 40
0010: e8 88 e6 0f 38 69 02 60 00 00 00 00 00 00 00 00
0020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0050: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0090: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00a0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00b0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00c0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00d0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00e0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00f0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0100: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0120: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0130: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0140: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0150: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0170: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0180: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0190: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
01a0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
01b0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
01c0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
01d0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
01e0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

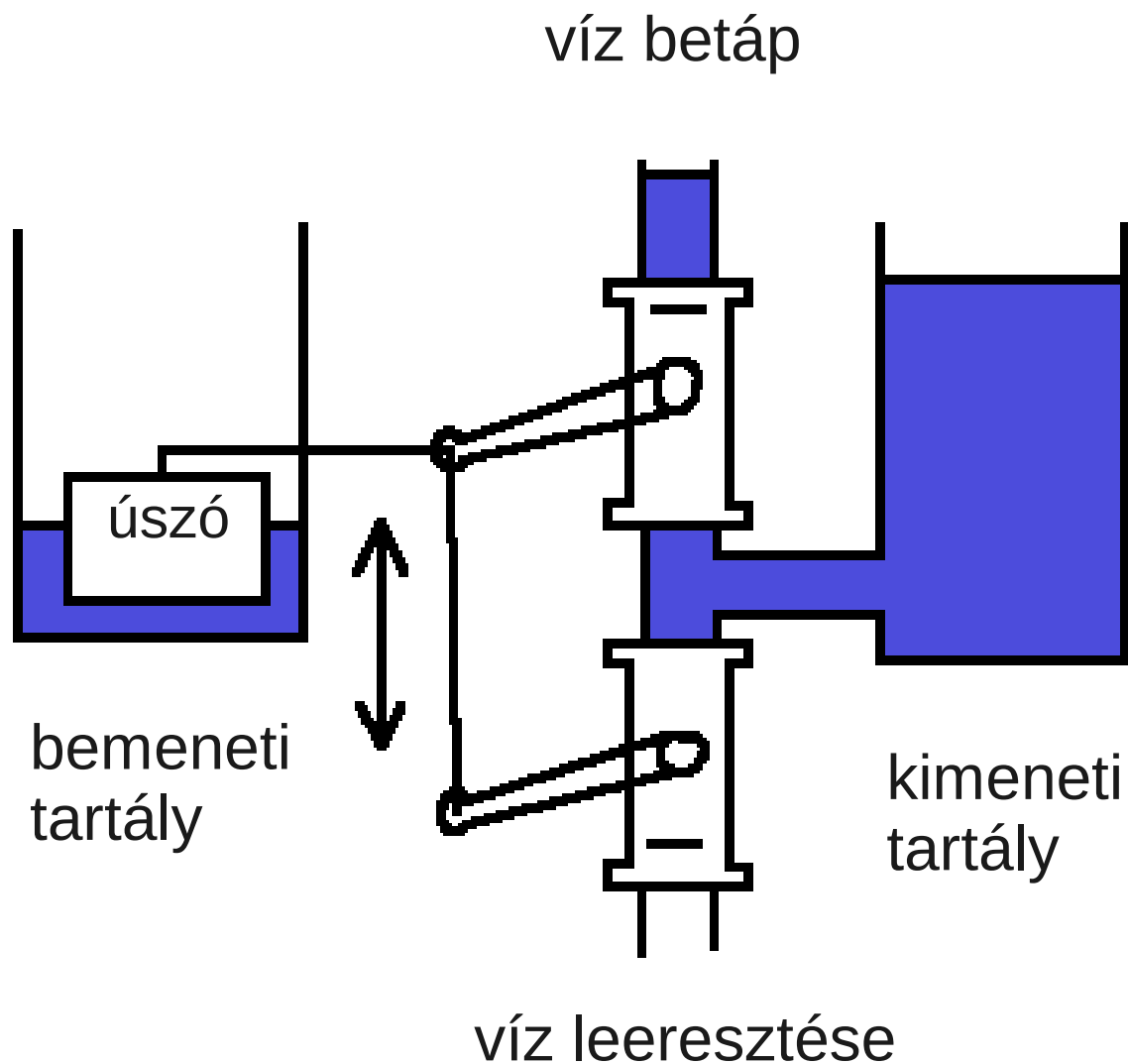
A víz analógia

elektromos áram	-	vízáram
áramerősség	-	vízhozam
feszültség	-	vízszint
ellenállás	-	szűkítő
tranzisztor	-	vízcsap
kondenzátor	-	tartály

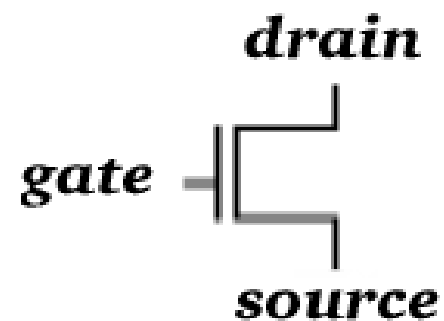
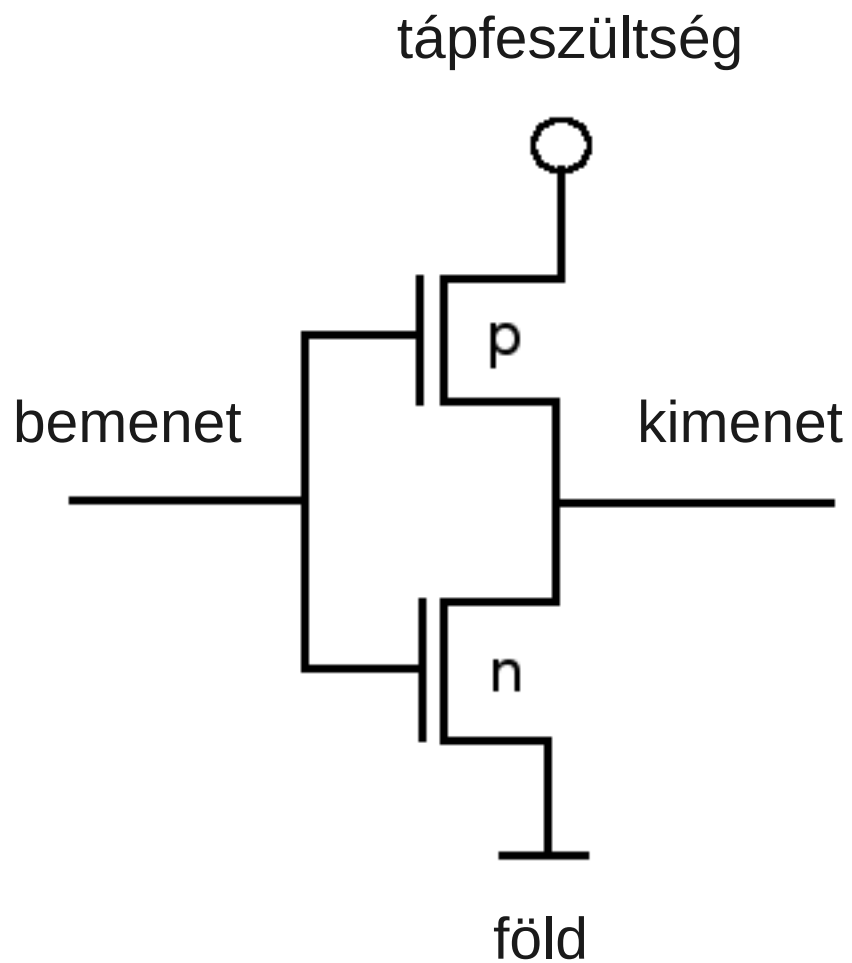
Az inverter

- A legegyszerűbb logikai kapu ($0 \rightarrow 1$, $1 \rightarrow 0$)
- Jelkésleltetés, oszcillátor
- Jelregenerálás
- Nagyáramú meghajtás
- Tárolók építése

Víz inverter CMOS módon

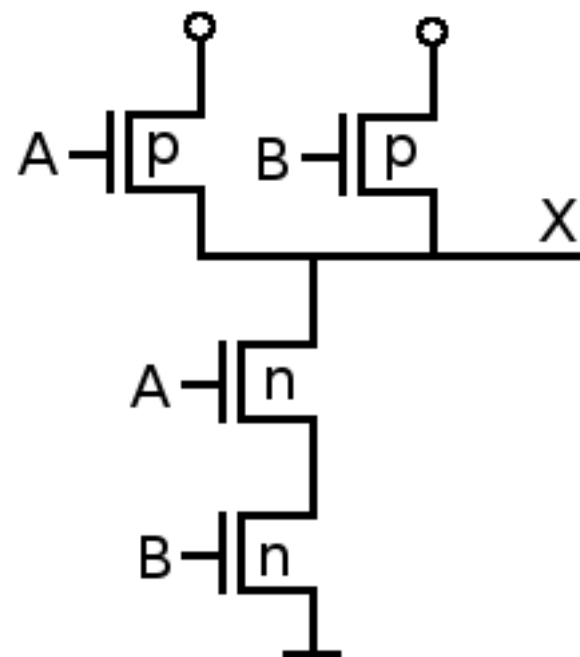
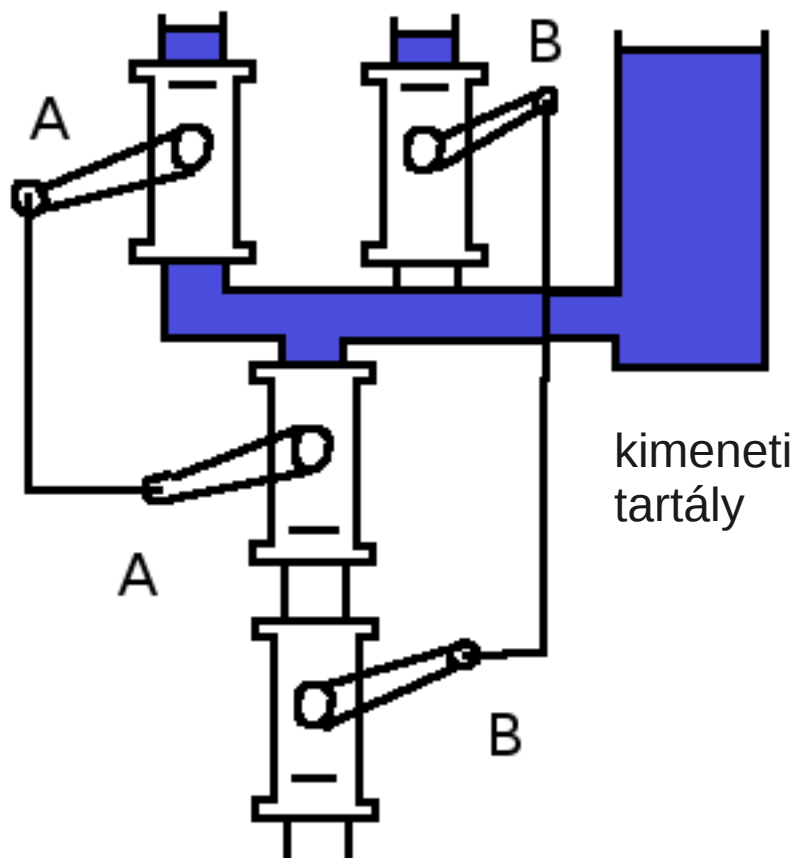


A valódi CMOS inverter



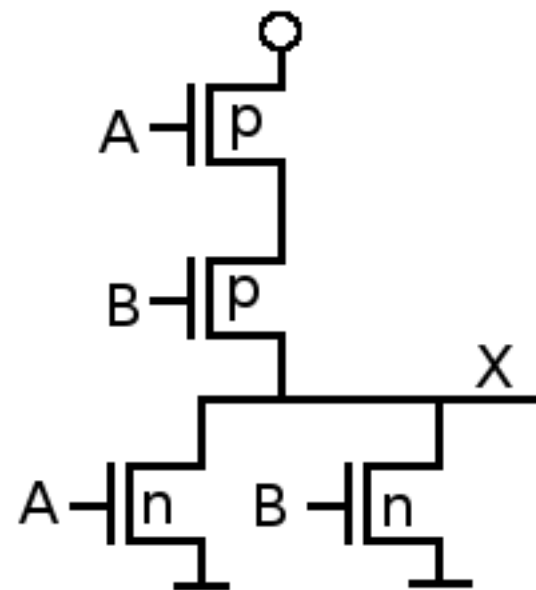
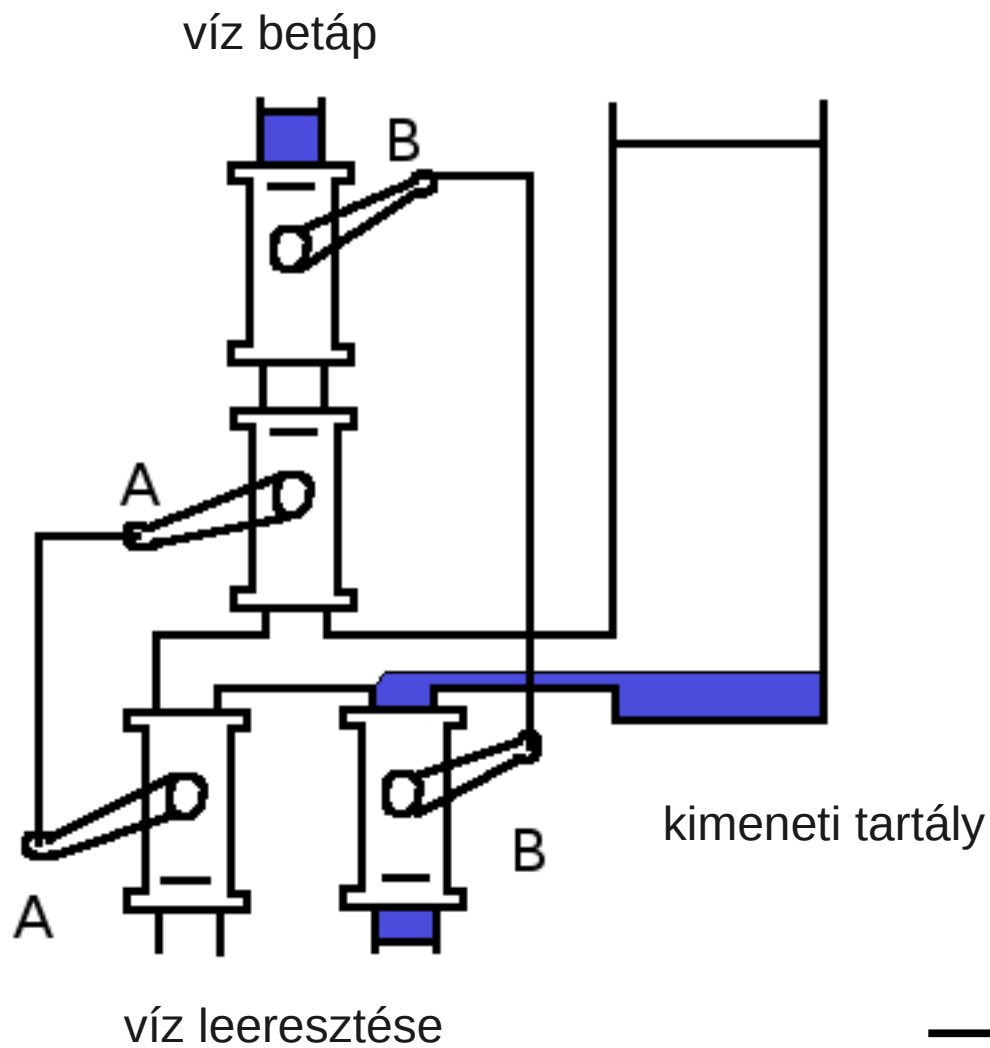
NAND kapu, CMOS

víz betáp



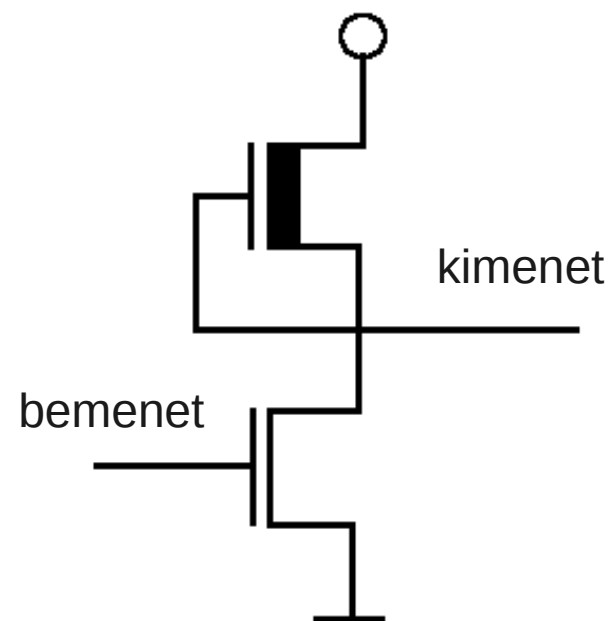
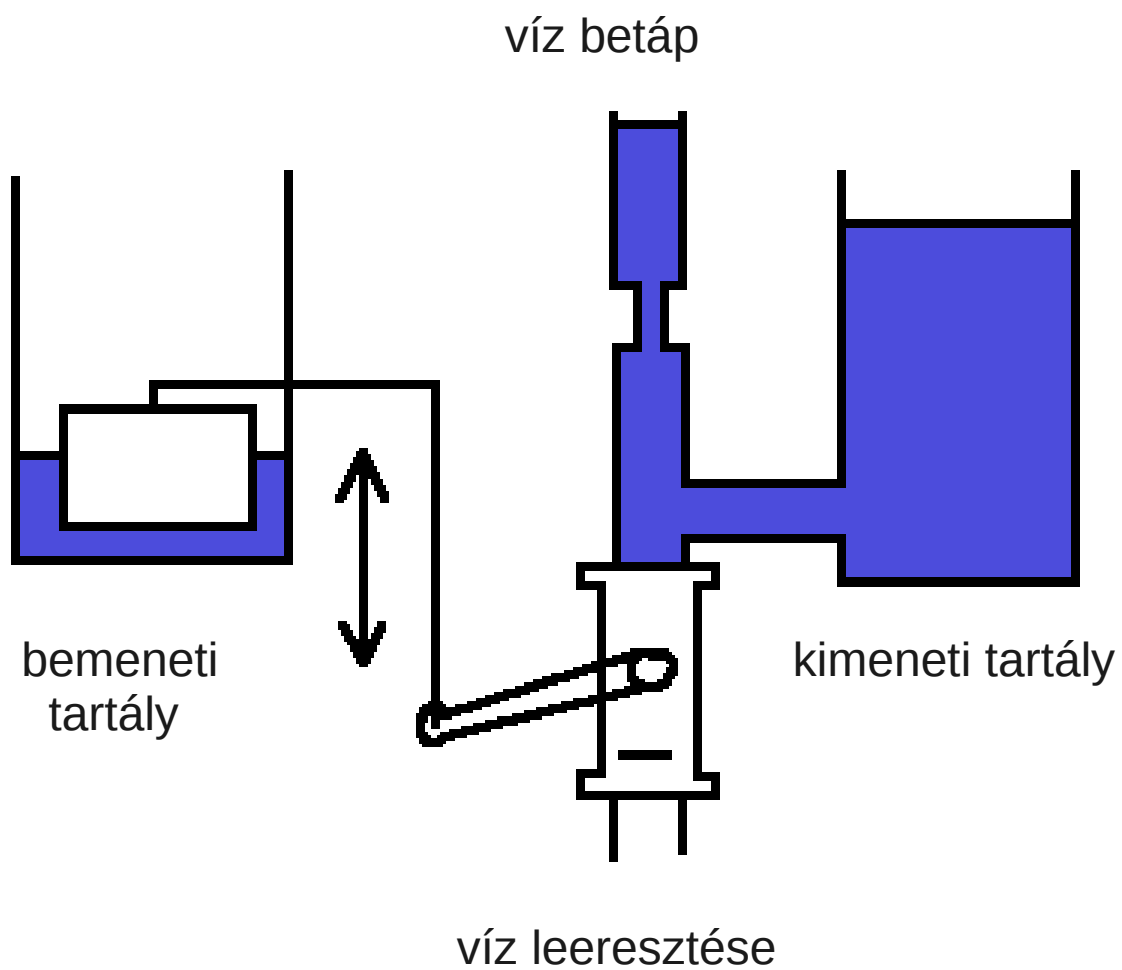
$$X = \overline{A * B}$$

NOR kapu, CMOS

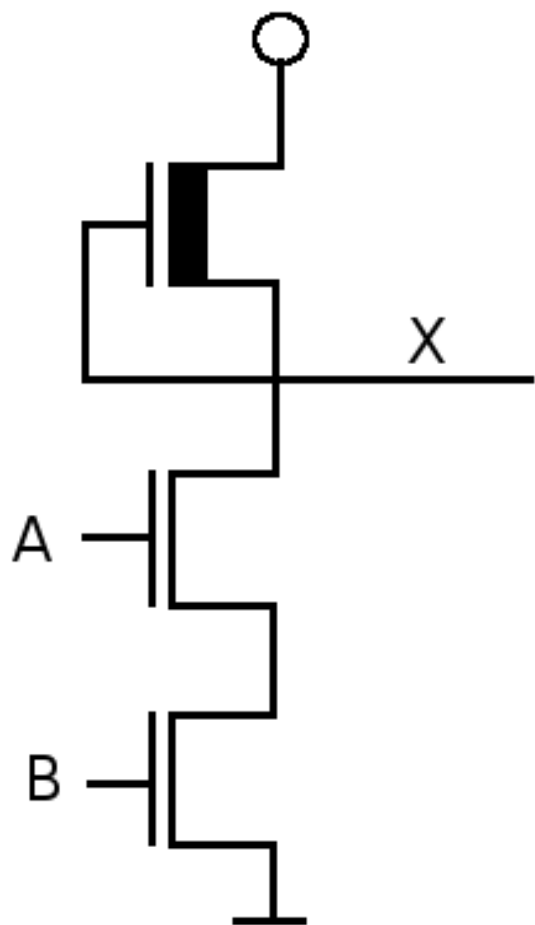


$$X = \overline{A + B}$$

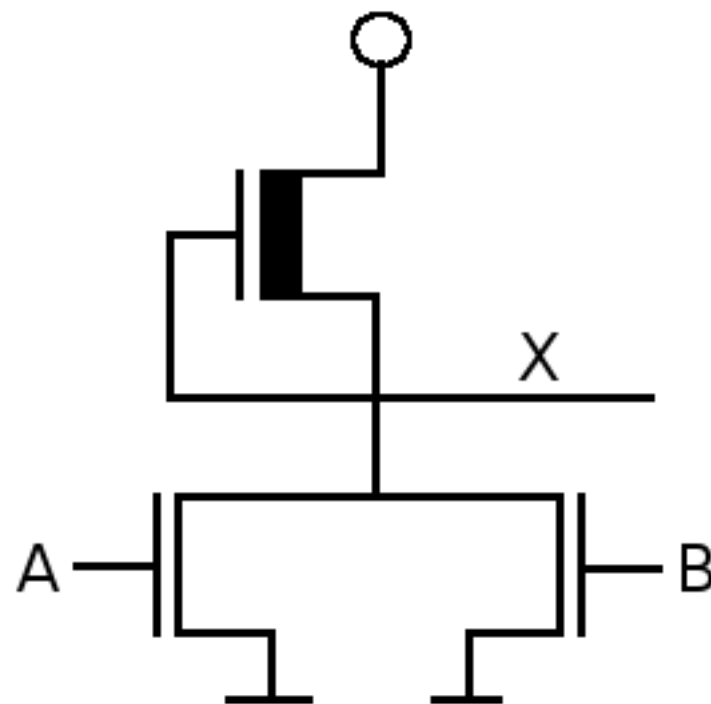
NMOS inverter



NMOS NAND és NOR

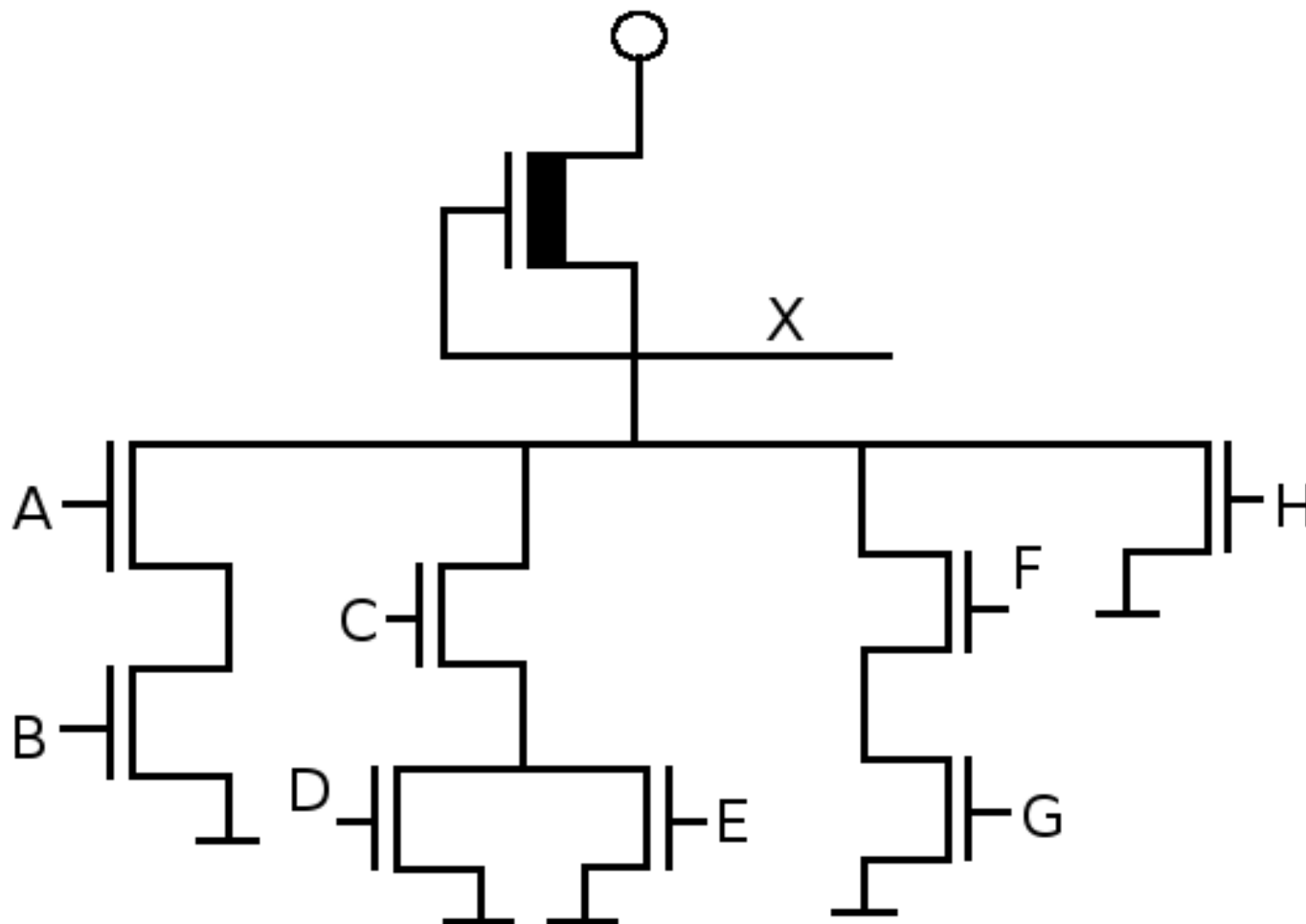


$$X = \overline{A * B}$$



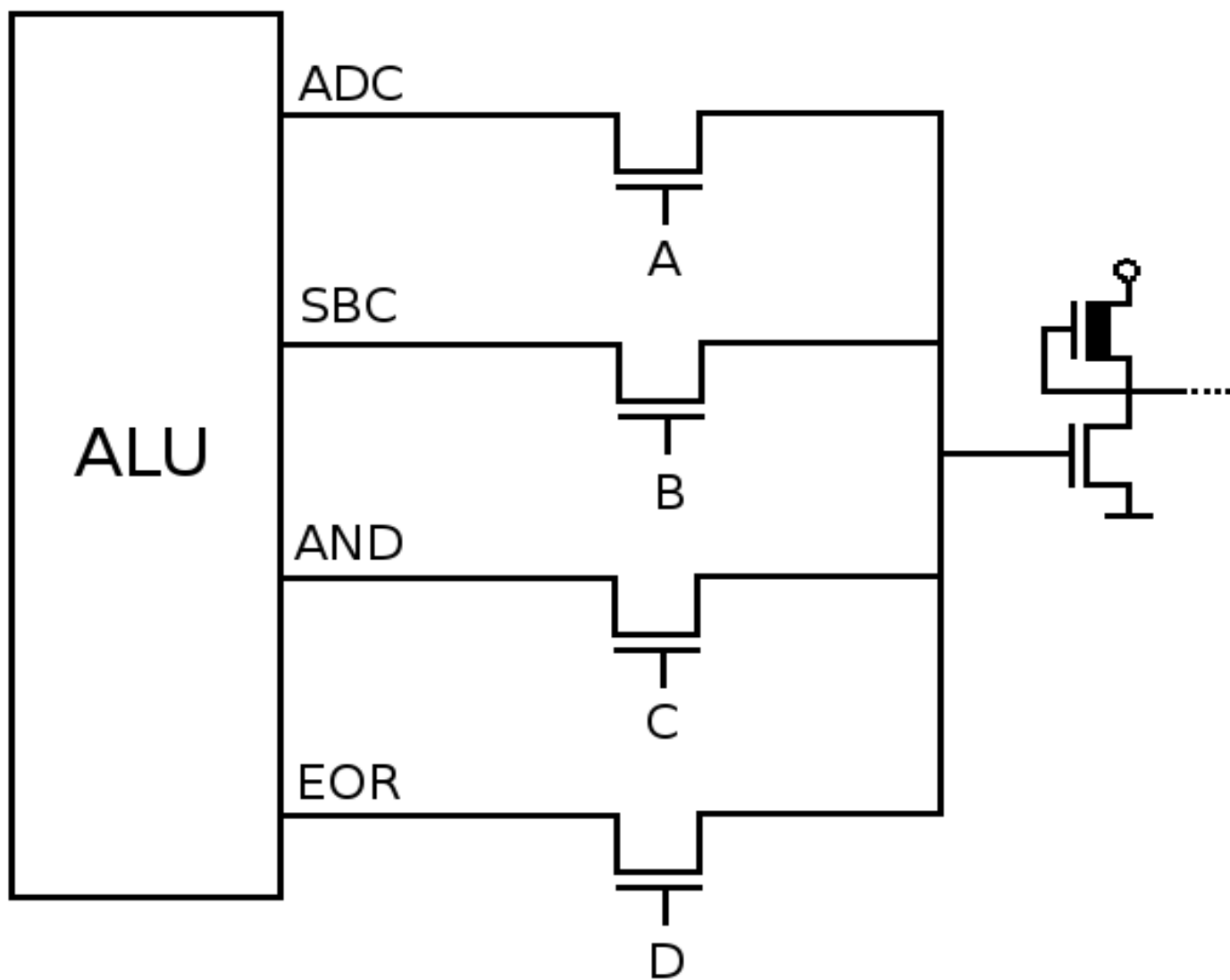
$$X = \overline{A + B}$$

NMOS komplex kapu

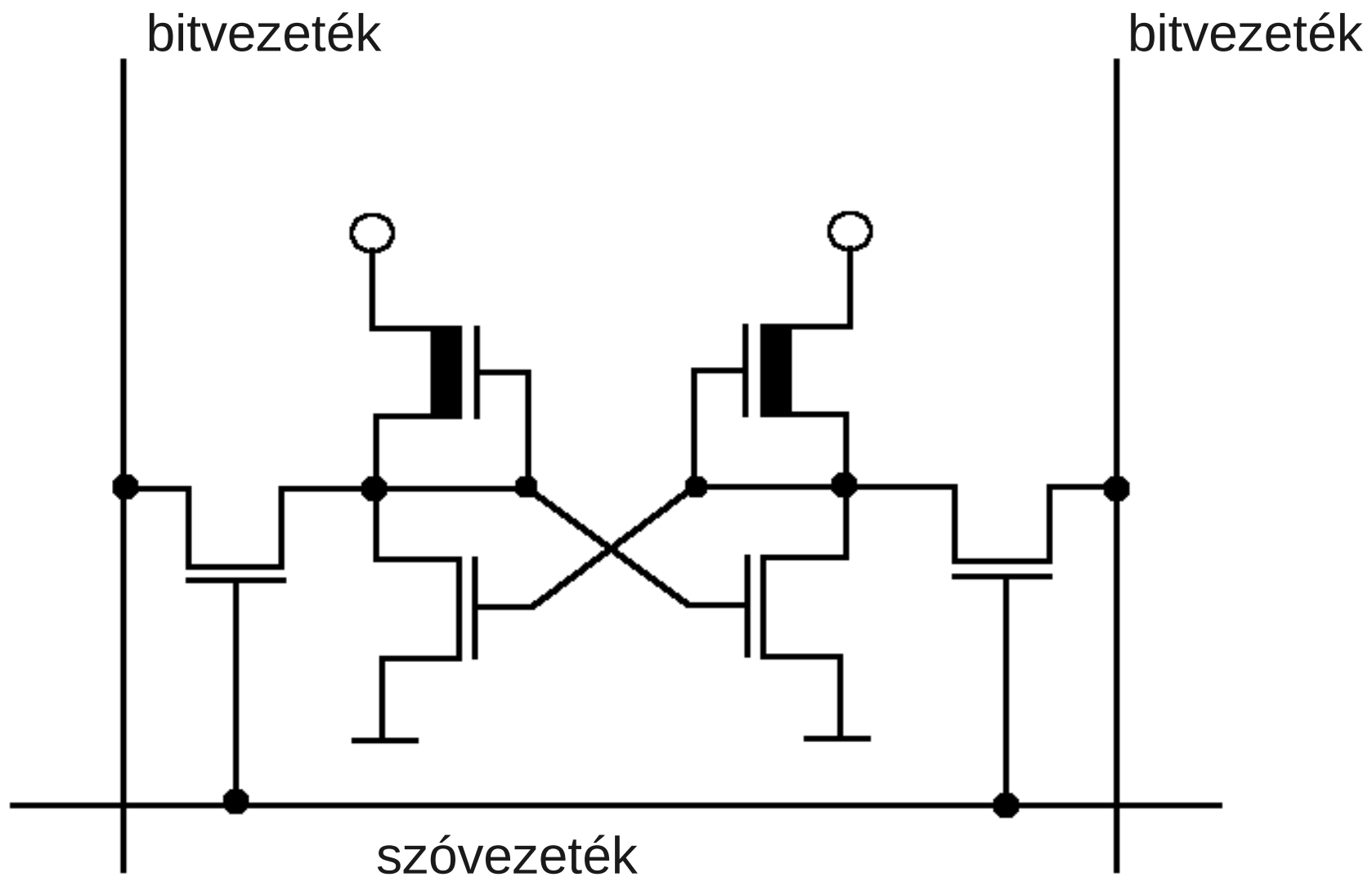


$$X = (A * B) + (C * (D + E)) + (F * G) + H$$

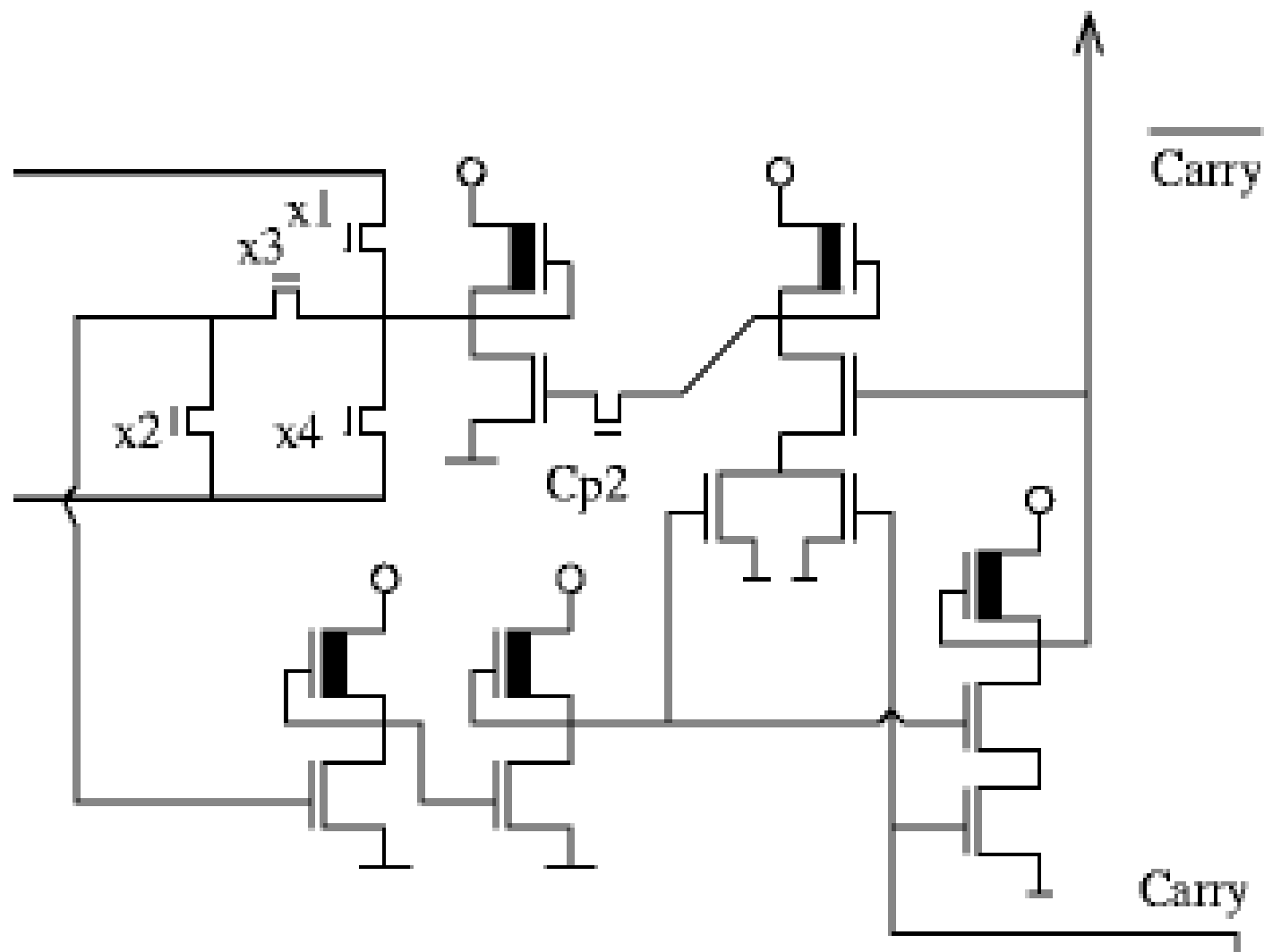
NMOS transfer gate



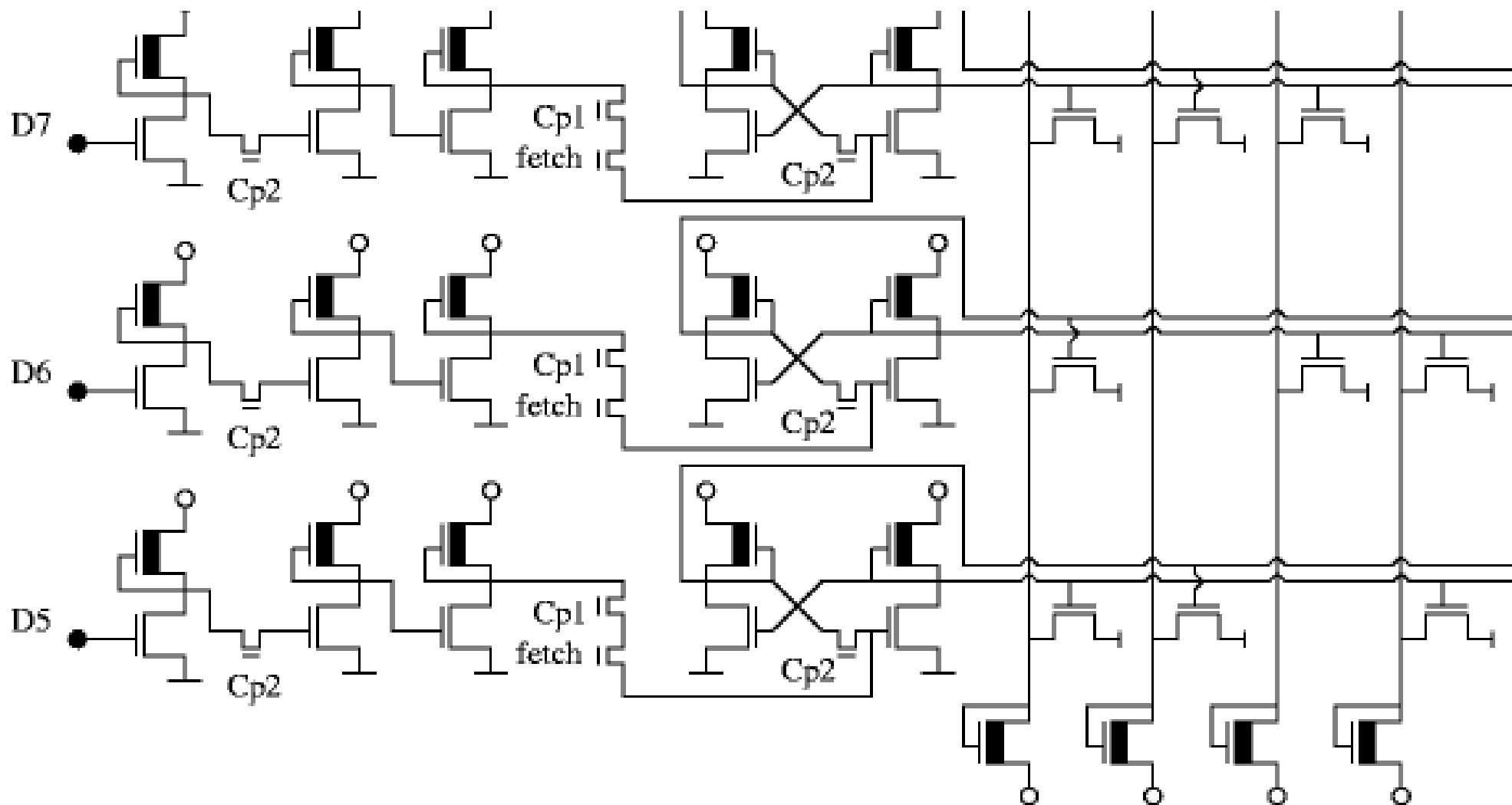
NMOS SRAM cella



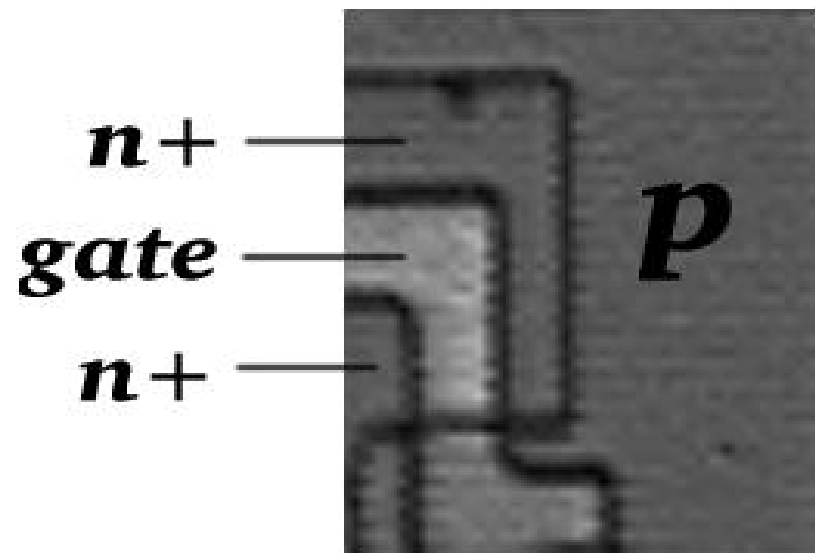
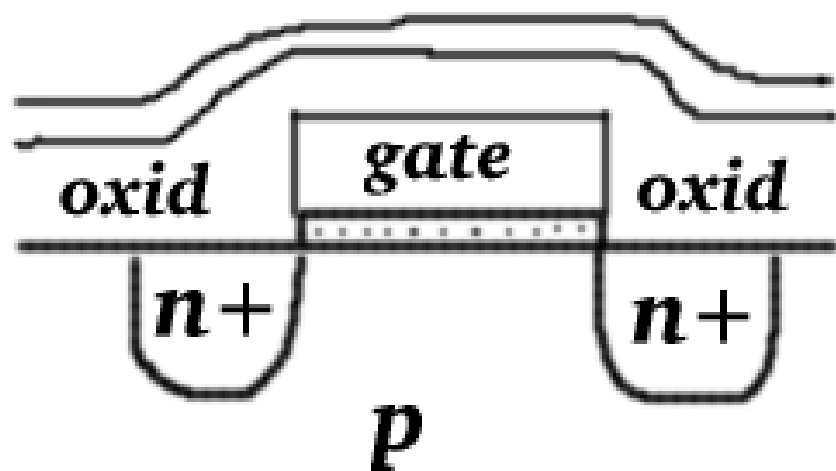
NMOS tároló komplex kapuval



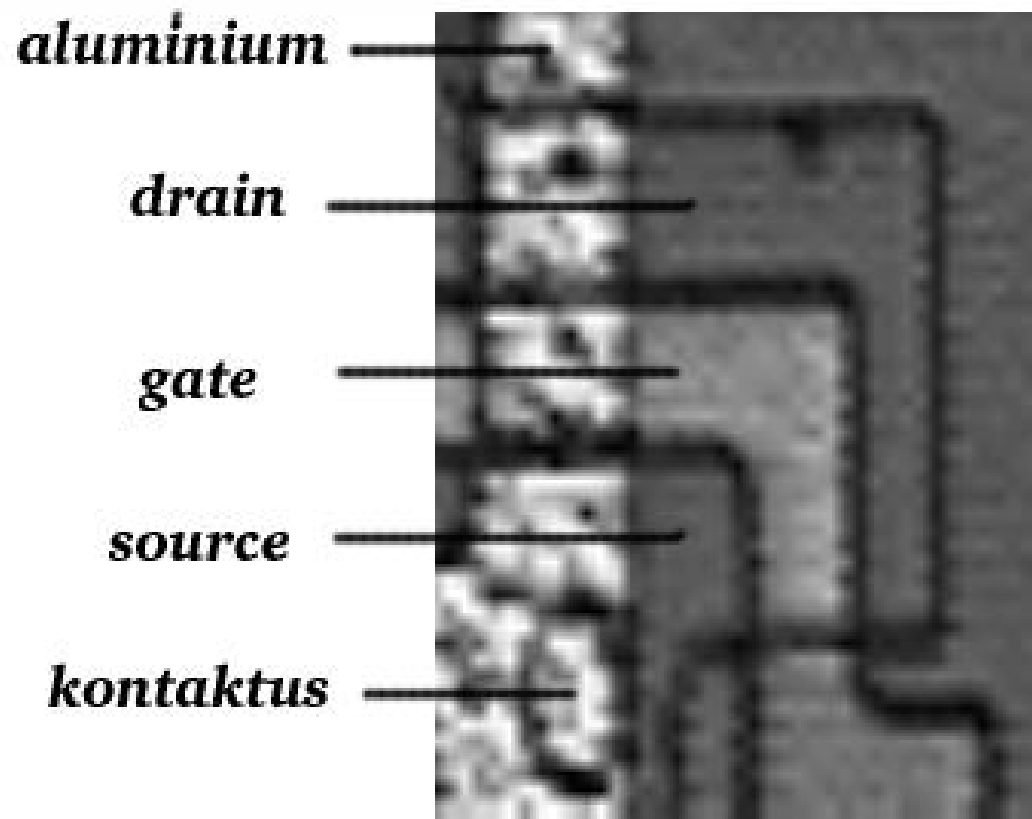
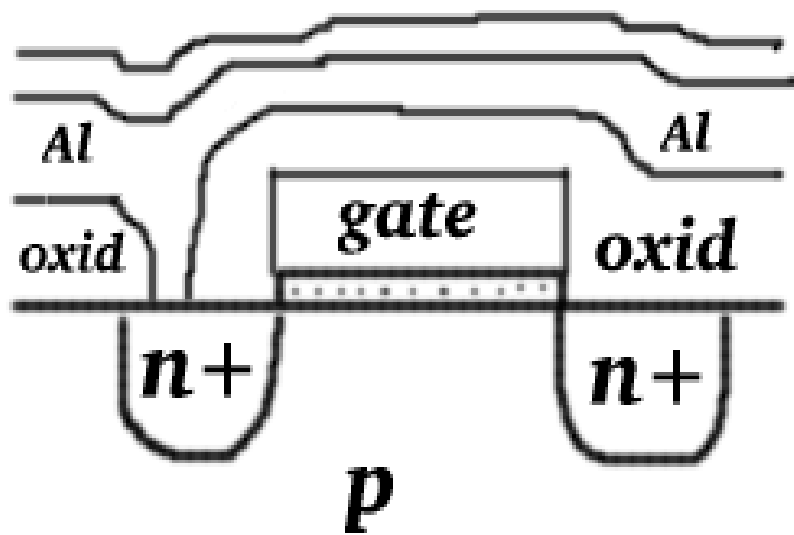
Utasításdekódoló



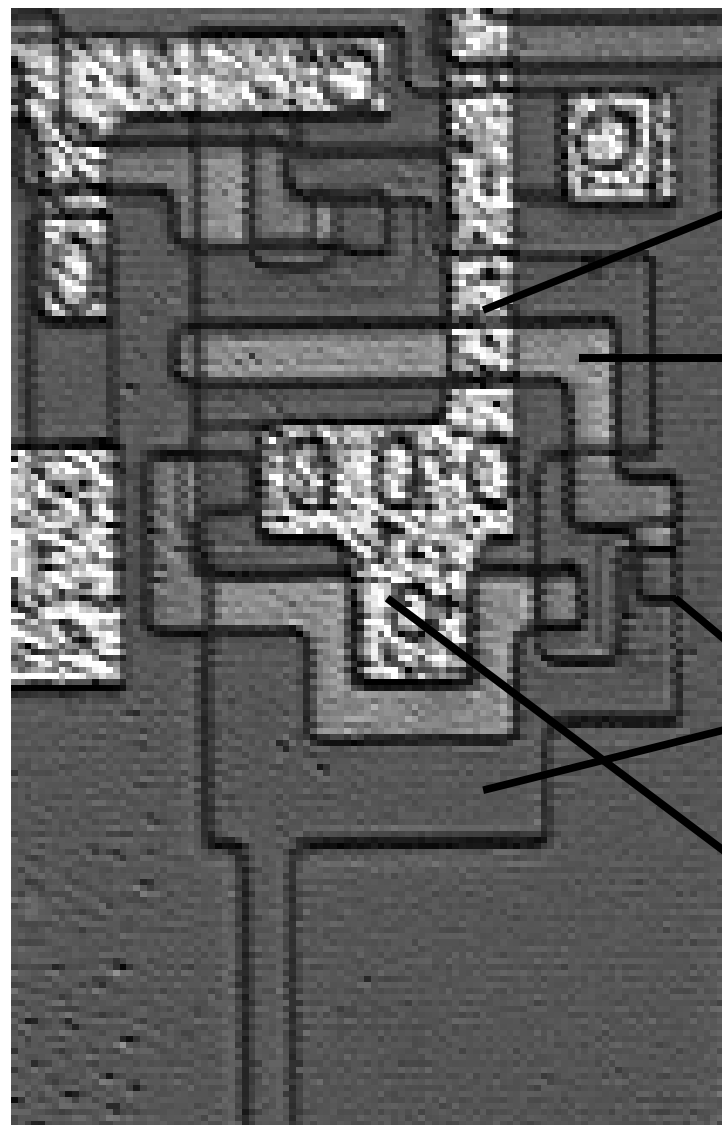
Tranzisztor és layout



Tranzisztor és layout, fémezéssel



Anyagok és funkciók



alumínium : vezetékelés

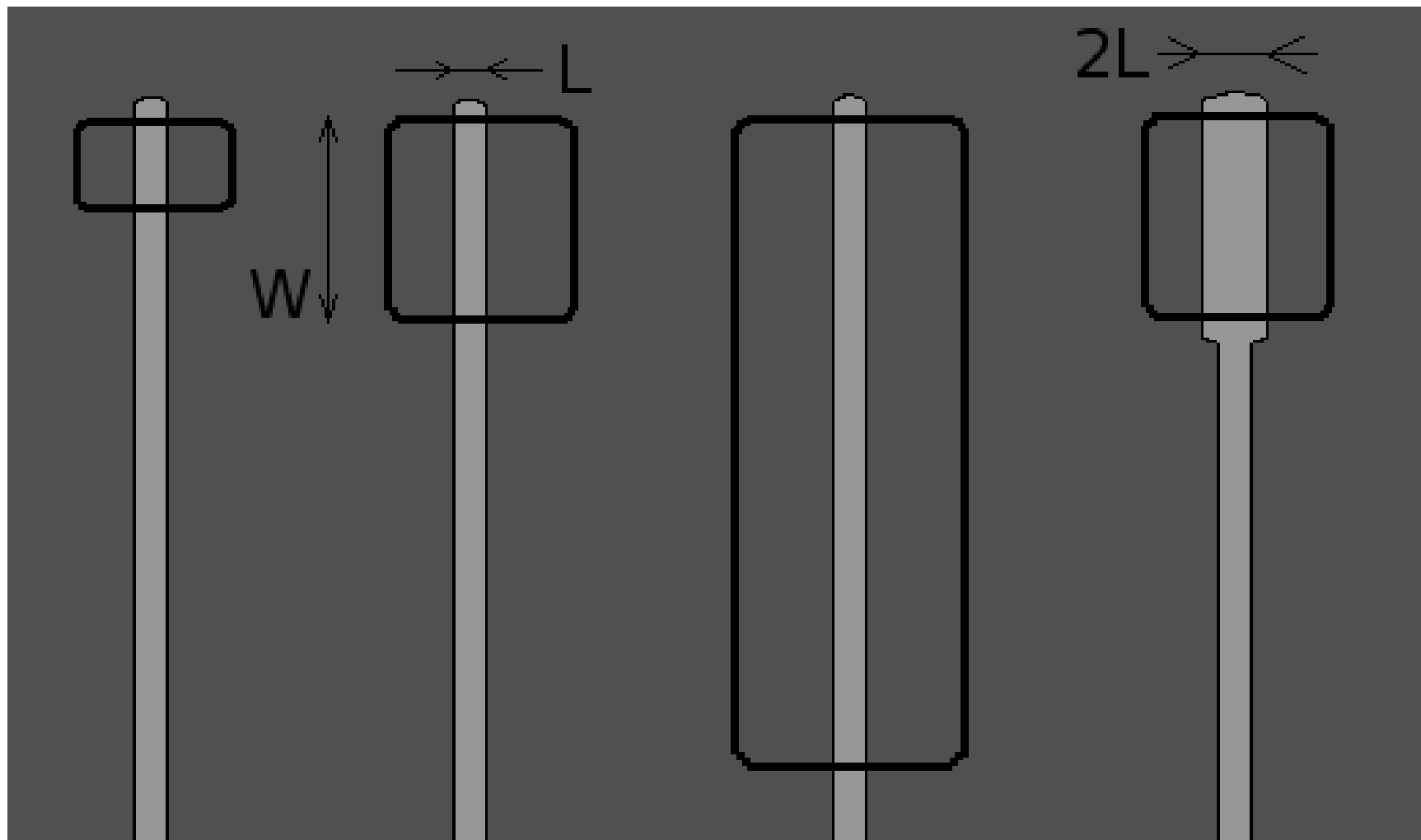
poliszilícium : vezeték és tranzisztor gate

aktív zóna : vezeték és tranzisztor
source/drain

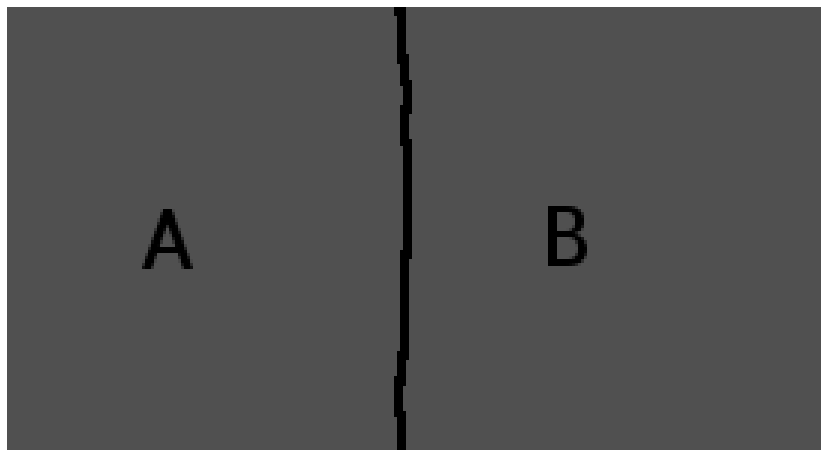
bújtatott kontaktus (poli és aktív)

kontaktus (alu és poli, alu és aktív)

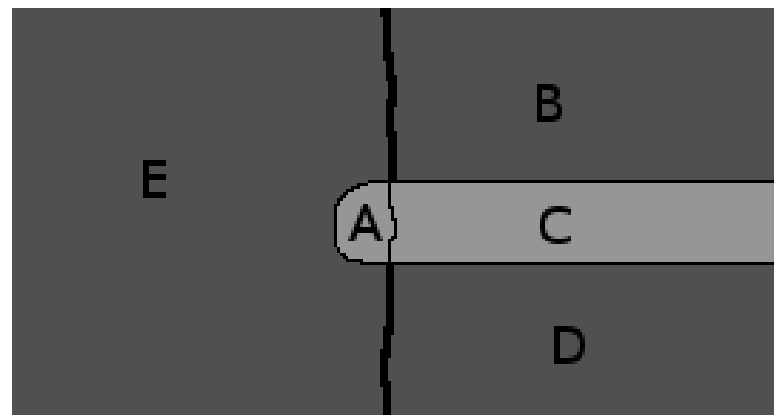
Csíkszélesség, W/L arány



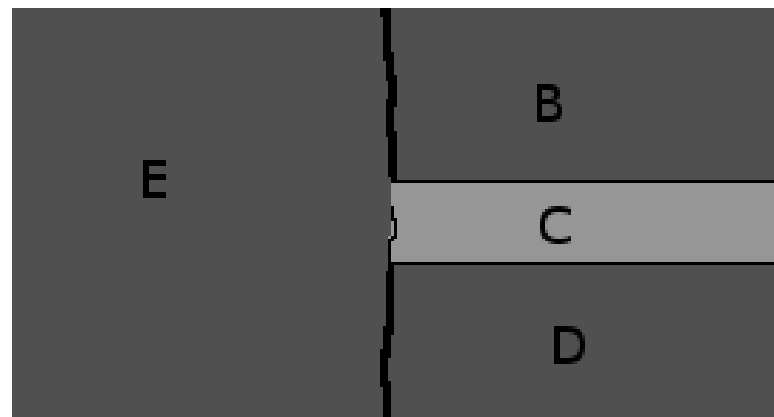
A gate túlnyúlás



a határvonal önmagában kevés

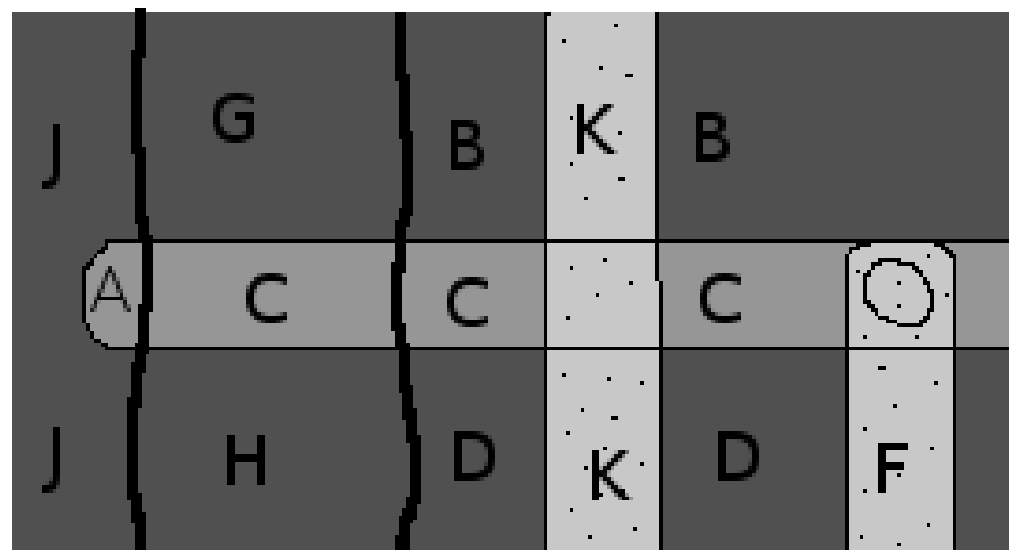
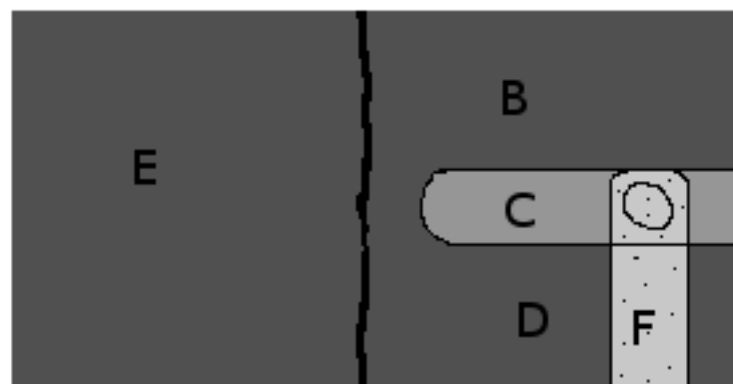


aktív zóna = B és D jelű terület



fiktív layout, nincs gate túlnyúlás

Gate túlnyúlás, fémezés

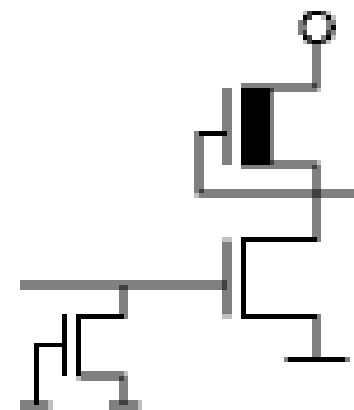
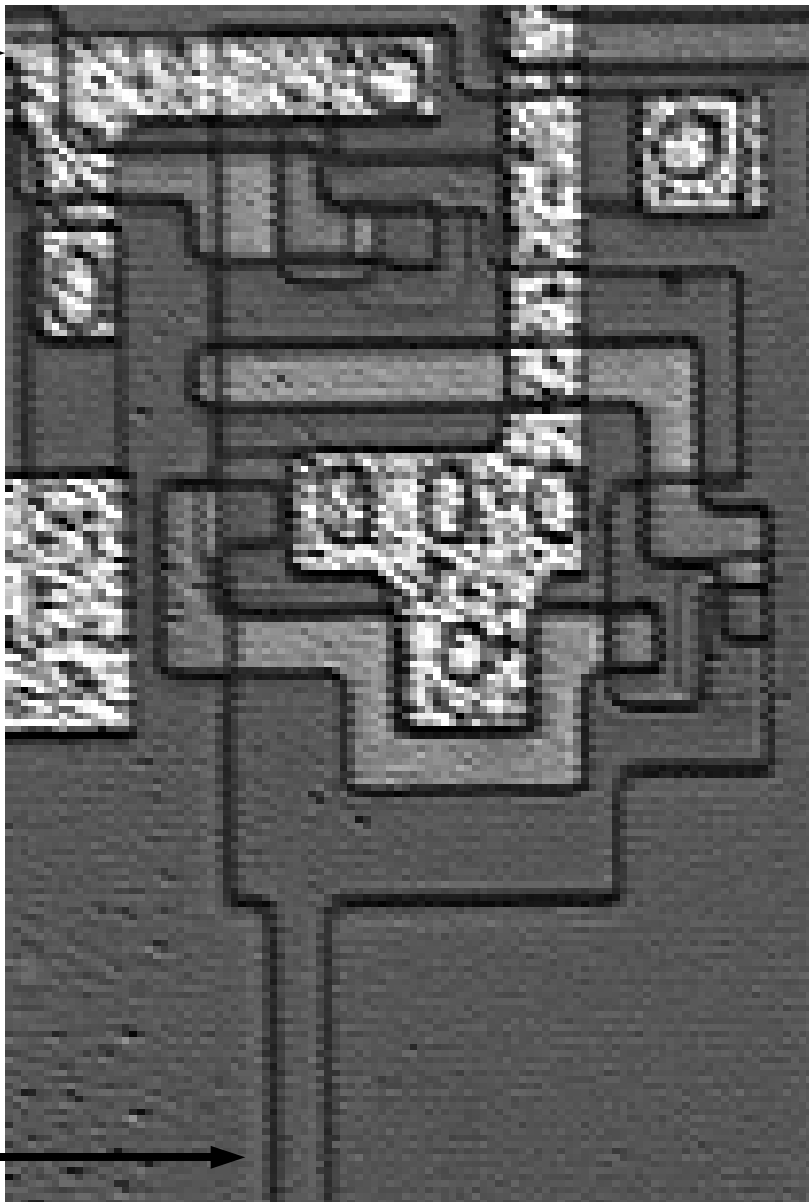


1. példa layoutfejtésre

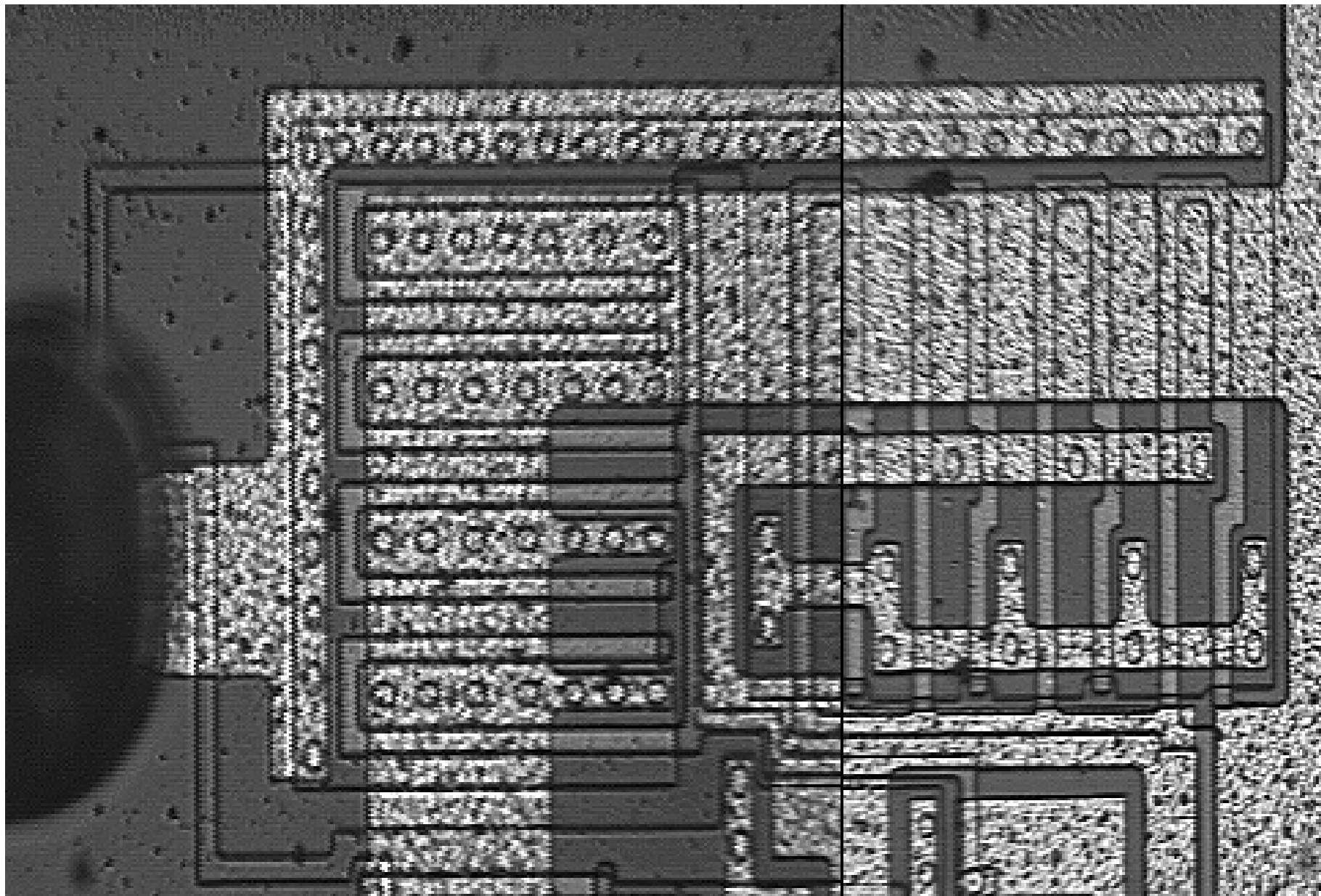
kimenet →

- 1.) aktív zóna, 3 tranzisztor
- 2.) terhelő tranzisztor: $2 * L$
- 3.) tápfeszültség, fém vezeték
- 4.) a másik fém a föld
- 5.) bemeneten lezárt tranzisztor
- 6.) bemenet -> bújtatott kontaktus
- 7.) driver tranzisztor

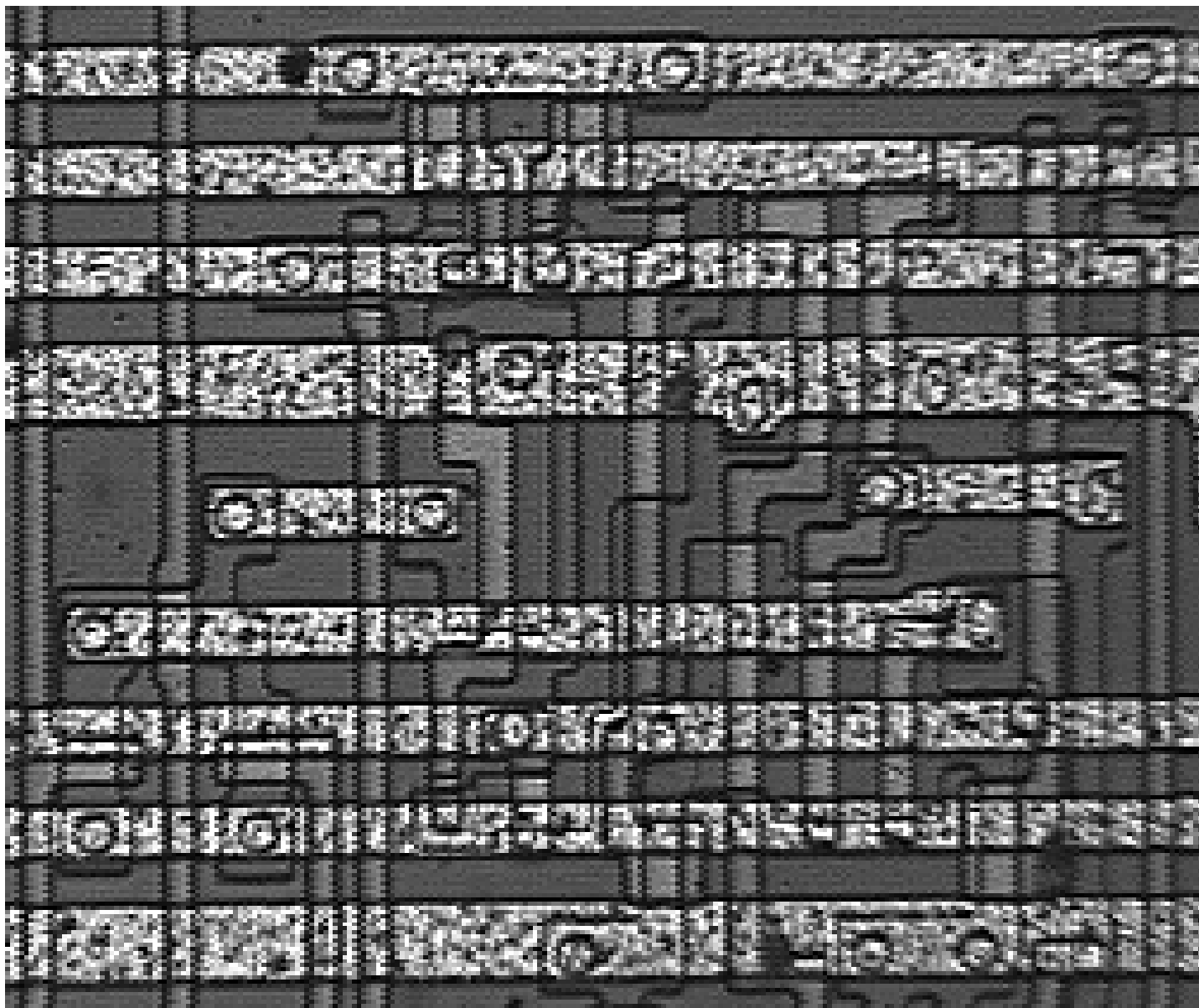
bemenet →



2. példa layoutfejtésre



3. példa layoutfejtésre



Köszönöm a figyelmet!

<http://impulzus.com/~bereg/hacktivity/>